

Assignment

Problem Statement

You are working on campaign management services that run different discounts for customers. These discounts can either be sponsored by the product group or via the vendors on the platform.

As a business owner, I want the following capabilities for discount management:

- I should be able to set discounts both on the overall cart and delivery separately.
- I should be able to run the discounts for x number of days and for my budget whatever gets reached first.
- A customer should only be able to use a discount only for x number of transactions in a day.
- I should be able to run the discount only for certain customers.

Requirements:

- Implement backend CRUD endpoints for campaign management.
- Provide a GET API endpoint to fetch available discount campaigns based on provided cart parameters.
- The backend must support the following discount capabilities:
 - Set discounts applicable either to the overall cart or specifically to delivery charges.
 - Enable campaign constraints by days or total budget (whichever limit is reached first).
 - Restrict discount usage to a specified maximum number of transactions per customer per day.
 - Allow campaigns to target specific customers.

Deliverables:

- API documentation clearly describing endpoints and request/response formats.
- Complete and testable backend code implementing the required functionality.
- Unit and integration tests covering key use cases and edge conditions(Preferred).

Technology Stack:

Python/FastApi/Django

Success Criteria:

- All listed requirements are fully implemented.
- The solution is robust, secure, and performant under expected traffic conditions.
- Clear and concise documentation provided.